2.Data Splitting and Cross Validation Program

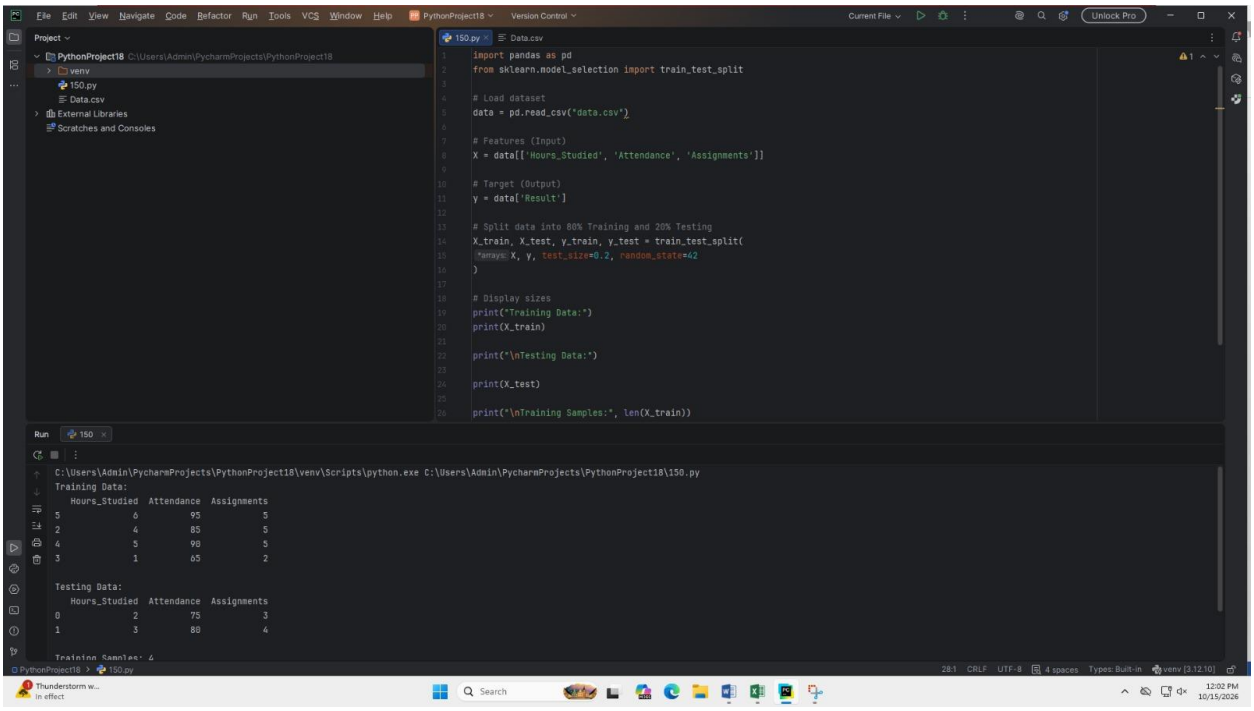
Aim :

To write a program that splits the given dataset into training and testing sets and then applies k-fold cross validation, so that the model's performance is checked on different parts of the data and not on a single split.

Algorithm :

1. Start the program.
2. Import the required libraries (pandas, train_test_split, KFold or cross_val_score from sklearn.model_selection, and a model such as LogisticRegression).
3. Load the dataset from the input file.
4. Separate the data into input features (X) and the target variable (y).
5. Split the data into training and testing sets using train_test_split (for example 80% training and
20% testing), with a fixed random_state.
6. Display the number of records in the training and testing sets.
7. Choose the number of folds k (for example k = 5) and create the k-fold object.
8. Divide the training data into k equal folds.
9. For each fold, use that fold as the validation set and the remaining k-1 folds as the training set.
10. Train the model on the training folds and calculate its accuracy on the validation fold.
11. Repeat steps 9 and 10 until every fold has been used once for validation.
12. Calculate the mean of the k accuracy scores to get the cross-validation score.
13. Display the score of each fold and the mean score.
14. Stop the program

Program&Result:


File Edit View Navigate Code Refactor Run Tools VCS Window Help
PythonProject18
Version Control
Current File
Unlock Pro
Project
PythonProject18 C:\Users\Admin\PycharmProjects\PythonProject18
venv
150.py
Data.csv
External Libraries
Scratches and Consoles
150.py
Data.csv
import pandas as pd
from sklearn.model_selection import train_test_split
Load dataset
data = pd.read_csv("data.csv")
Features (Input)
X = data[['Hours_Studied', 'Attendance', 'Assignments']]
Target (Output)
y = data['Result']
Split data into 80% Training and 20% Testing
X_train, X_test, y_train, y_test = train_test_split(
*arrays: X, y, test_size=0.2, random_state=42
)
Display sizes
print("Training Data:")
print(X_train)
print("\nTesting Data:")
print(X_test)
print("\nTraining Samples:", len(X_train))
Run
150
C:\Users\Admin\PycharmProjects\PythonProject18\venv\Scripts\python.exe C:\Users\Admin\PycharmProjects\PythonProject18\150.py
Training Data:
Hours_Studied Attendance Assignments
5 6 95 5
2 4 85 5
4 5 98 5
3 1 65 2
Testing Data:
Hours_Studied Attendance Assignments
0 2 75 3
1 3 80 4
Training Samples: 4
PythonProject18 > 150.py
28:1 CRLF UTF-8 4 spaces Types: Built-in venv (3.12.10)
Thunderstorm w...
In effect
Search
12:02 PM
10/15/2026